Student Social Media & Mental Health Analytics

Power BI Capstone Project — Project Documentation

5,000+	5	18	6	5
Student Records	Dashboard Pages	Dataset Columns	KPI Cards	DAX Measures

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Project Type	Power BI Masterclass — Capstone Project
Dataset	Student Social Media & Mental Health Impact (5,000 rows)
Tool Used	Microsoft Power BI Desktop
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Power BI Masterclass · Infinity Code Nexus · Education Analytics · Mental Health Analytics · Behaviour Analytics

1. Executive Summary

This Power BI capstone project investigates the relationship between social media usage patterns and student mental health outcomes across a global cohort of **5,000 students**. Using Microsoft Power BI Desktop, an interactive five-page analytical dashboard was designed and built from a single cleaned dataset containing 18 fields spanning demographics, digital behaviour, sleep, stress, academic performance, and mental wellbeing.

The dashboard surfaces six critical KPIs and provides drill-down analysis across four analytical lenses — social media behaviour, mental health, lifestyle, and country-level performance. Key findings confirm a measurable inverse relationship between daily social media consumption and student mental health scores, with students in the Very High usage category averaging significantly lower wellbeing scores than their Low usage counterparts.

Dashboard KPI Snapshot

5,000	6.23	5.08	2.81	6.63	3.01
Total Students	Avg Mental Health	Avg Daily Usage (hrs)	Avg Stress (1-4)	Avg Sleep (hrs)	Avg Study (hrs)

2. Business Problem Statement

Student mental health has become a critical concern for educational institutions worldwide. The exponential growth of social media platforms—particularly entertainment-focused applications such as TikTok and Instagram—has raised questions about their impact on academic performance, sleep quality, stress levels, and overall psychological wellbeing.

Problem 1	Do students who spend more hours on social media report lower mental health scores?
Problem 2	Which social media platforms correlate with the highest student stress levels?
Problem 3	How does sleep quality mediate the relationship between stress and mental health?
Problem 4	Do mental health outcomes differ significantly across academic levels and countries?
Problem 5	Is study time being displaced by social media usage among students?

3. Dataset Overview

The dataset *Student_Social_Media_And_Mental_Health_Impact.csv* contains **5,000 rows** and was enriched to **18 columns** after Power Query transformations. All columns passed validation with 100% valid, 0% error, and 0% empty values.

#	Column Name	Type	Description
1	Age	Integer	Student age (17–35)
2	Gender	Text	Male / Female / Non-binary
3	Country	Text	Country of residence (geocoded for map)
4	Academic_Level	Text	High School / Undergraduate / Graduate
5	Most_Used_Platform	Text	Primary social media platform (12 platforms)
6	Purpose_Of_Use	Text	Education / Entertainment / Networking / News
7	Avg_Daily_Usage_Hours	Decimal	Average daily screen time (0.5–9.0 hrs)
8	Daily_Unlocks	Integer	Number of phone unlocks per day (10–300)
9	Study_Hours	Decimal	Average daily study time (0–12 hrs)
10	Physical_Activity_Hours	Decimal	Daily physical activity (0–5 hrs)
11	Sleep_Hours_Per_Night	Decimal	Sleep duration (3–10 hrs)
12	Stress_Level	Text	Self-reported: Low / Medium / High / Very High
13	*Mental_Health_Score*	*Decimal*	*Wellbeing score 1–10 (higher = better)*
—	*Usage_Category*	*Calc.*	*Screen time band: Low / Moderate / High / Very High*
—	*Sleep_Category*	*Calc.*	*Sleep quality: Good (≥8h) / Fair (≥6h) / Poor (<6h)*
—	*Age_Group*	*Calc.*	*Teen / Young Adult / Adult / Older Adult*
—	*Study_vs_Social*	*Calc.*	*Study Hours – Avg Daily Usage Hours*
—	*Stress_Order*	*Calc.*	*Numeric sort key: Low=1, Medium=2, High=3, VH=4*

Rows marked '—' in the # column are Power Query calculated columns added during data transformation.

4. Technical Architecture

4.1 Data Pipeline

Stage	Tool / Layer	Actions Performed
Ingestion	Power Query	CSV import via Get Data → Text/CSV; Transform Data selected
Cleaning	Power Query	Data type fixes, null handling, whitespace trim, capitalization
Transformation	Power Query (M)	5 calculated columns added via Add Column → Custom Column
Modelling	Power BI Model View	Stress Level lookup table; Country geocoded; _Measures table
Measures	DAX	6 KPI measures + 3 conditional colour measures + 2 advance
Visualization	Power BI Canvas	5 dashboard pages; 4 synced slicers; conditional formatting

4.2 Data Model

The model uses a **star-schema lite** design with three tables:

Table	Role	Key Columns
StudentData	Fact table (5,000 rows × 18 cols)	All raw + calculated columns
Stress I	Dimension — Stress Level sort order (1:*)	Stress_Level, Sort_Order
_Measures	Measures table (no data rows)	Hosts all 11 DAX measures

4.3 Core DAX Measures

Measure Name	DAX Pattern	Purpose
Total Students	COUNTROWS	Count of all student records
Avg Mental Health Score	AVERAGE	Primary wellbeing KPI
Avg Daily Usage Hours	AVERAGE	Screen time indicator
Avg Study Hours	AVERAGE	Academic effort metric
Avg Sleep Hours	AVERAGE	Lifestyle wellbeing metric
Avg Stress Numeric	AVERAGEX + SWITCH	Text stress converted to 1–4 numeric scale
High Stress %	DIVIDE + FILTER	% students with High or Very High stress
Good Sleep %	DIVIDE + FILTER	% students sleeping ≥7 hours/night
Mental Health Color	IF (conditional)	Returns HEX colour for conditional formatting
Stress Color	SWITCH(TRUE())	Returns HEX colour based on stress level
Usage Risk Color	IF (conditional)	Returns HEX colour based on usage band

5. Dashboard Pages

Page 1 — Executive Overview

- 6 KPI cards: Total Students, Avg Mental Health, Avg Daily Usage, Avg Study, Avg Sleep, Avg Stress
- World map: Avg Mental Health Score bubble size by Country
- Stress Level Distribution horizontal bar chart (colour-coded Low → Very High)
- Platform Distribution donut chart (12 platforms)
- 4 synced slicers: Country, Gender, Most_Used_Platform, Academic_Level
- Clear All slicer reset button

Page 2 — Social Media Analytics

- Most Used Social Media Platform bar chart (Instagram leads: 1,130 students)
- Platform Usage by Purpose stacked bar — Education vs Entertainment vs Networking vs News
- Avg Daily Unlocks by Usage Category column chart (Very High: 230+ unlocks/day)
- All 4 global slicers synced

Page 3 — Mental Health Analytics

- Stress Level Distribution pie chart (Very High 32.4%, High 28.8%)
- Mental Health Heatmap: matrix of Stress Level × Sleep Category with conditional colour
- Avg Mental Health Score by Academic Level bar (Graduate 6.4 > UG 6.2 > High School 5.9)
- Scroll-based slider for detailed drill-in

Page 4 — Lifestyle Analytics

- Sleep Quality Distribution combo chart: student count + Avg Mental Health Score per sleep band
- Study Hours vs Social Media Usage by Academic Level grouped bar
- Physical Activity vs Mental Health Score line chart by Gender (positive correlation visible)
- All groups show social media usage exceeding study hours

Page 5 — Country Analytics

- Multi-metric world map: Total Students, Avg Stress, Avg Sleep, Avg Daily Usage in tooltips
- Country Performance Ranking table with conditional formatting on mental health column
- Avg Mental Health Score by Country bar chart (Malta 8.4 leads; top countries in Eastern Europe)
- Academic Level count by Country horizontal bar (India 389, USA 355, Canada 230)

6. Key Insights from Data

01 Usage & Mental Health Correlation

Students averaging >5 hours of daily social media use (avg: 5.08 hrs across all cohorts) report an average mental health score of 6.23/10. Very High usage category students consistently show the lowest wellbeing scores, confirming the negative correlation. Daily Unlocks further reinforces this — Very High users unlock phones 230+ times daily.

Action: Introduce digital wellness campaigns targeting students with >5 hrs/day usage.

02 Platform Risk Profile

Instagram dominates usage with 1,130 students (22.6%) followed by TikTok (918, 18.4%). Both are primarily entertainment-driven platforms. LinkedIn and YouTube, which score higher on Education purpose-of-use, are used by comparatively fewer students. Entertainment-focused platforms correlate with higher stress and lower sleep quality.

Action: Educational institutions should promote academic use of LinkedIn and YouTube as productive digital alternatives.

03 Stress Distribution is Alarming

A combined 61.2% of the student population reports High or Very High stress (Very High: 32.4%, High: 28.8%). Only 12.9% report Low stress. The Mental Health Heatmap confirms that Very High Stress + Poor Sleep produces the lowest mental health scores (<5.0), while Low Stress + Good Sleep exceeds 7.8.

Action: Universities should deploy targeted counselling outreach for the 32.4% Very High stress cohort, prioritising sleep hygiene as a first intervention.

04 Sleep is a Critical Mediator

Average sleep across all students is 6.63 hrs/night — below the recommended 8 hours. The Sleep Quality chart reveals the majority fall into the 'Fair' sleep band (6–7.9 hrs). 'Good' sleepers (≥8 hrs) represent the smallest group but show substantially higher mental health scores. The heatmap shows Good Sleep raises scores across all stress levels.

Action: A 'No screens 1 hour before bed' awareness campaign could shift students from Fair to Good sleep quality, with measurable mental health benefits.

05 Social Media Displacing Study Time

Average daily social media usage (5.08 hrs) exceeds average study hours (3.01 hrs) by 2.07 hours — a 69% displacement gap. The Lifestyle page chart shows this pattern holds across all academic levels: High School, Undergraduate, and Graduate students all spend more time on social media than studying.

Action: Embed digital time-management tools and study-vs-usage comparison dashboards into student portals to increase self-awareness.

06

Academic Level & Mental Health

Graduate students score highest on mental health (6.4) followed by Undergraduates (6.2) and High School students (5.9). High School students have the lowest scores despite lower average study hours, suggesting environmental and social pressures rather than academic workload alone drive stress at this level.

Action: High School-level mental health programmes deserve priority — academic pressure alone does not explain the gap; social comparison and platform addiction are likely factors.

7. Business Recommendations

Based on the analytical findings above, the following stakeholder-specific recommendations are proposed. Each recommendation follows the Observation → Insight → Action framework.

For Universities & Colleges

→ Launch Digital Wellness Programmes targeting students averaging >5 hrs/day. Create peer-led screen time awareness campaigns that tie usage to measurable GPA impact.

→ Embed platform purpose guidance in student orientation — promote LinkedIn for professional networking and YouTube for educational content as alternatives to passive TikTok/Instagram scrolling.

→ Introduce mandatory mental health check-in surveys at semester start and mid-term, filtered by academic level to target the High School cohort (lowest avg MH: 5.9).

For Student Counsellors

→ Prioritise sleep hygiene workshops — students sleeping <6 hrs show the worst combination of high stress and low mental health scores. Simple interventions: 'Phone curfew 10pm' campaigns can shift Fair → Good sleep for a measurable score improvement.

→ Use the Stress Level data to identify and proactively contact the Very High stress cohort (32.4% of students) for early counselling sessions before exam periods.

→ Physical activity programmes should be promoted; the Lifestyle page confirms a clear positive correlation between physical activity hours and mental health score across genders.

For Policy Makers

→ Partner with social media platforms to introduce student-mode usage caps (e.g. 2 hr/day) with parental/institutional dashboard visibility for under-18 students.

→ Fund longitudinal research to establish causal pathways — the current dataset shows strong correlational evidence; establishing causality requires tracking the same cohort across multiple academic years.

→ Review national guidelines for screen time in school curricula and embed digital literacy education, focusing on platform purpose awareness (entertainment vs education).

For Researchers

→ Expand dataset scope to include semester-level academic performance (GPA) to quantify the exact academic cost of excessive social media usage.

→ Investigate the country-level variance — Malta (8.4), Chile (8.1) and Iraq (8.1) lead the mental health rankings despite varying usage levels. Cultural and educational system factors likely mediate outcomes and deserve dedicated study.